



Dwight Look College of

ENGINEERING
TEXAS A&M UNIVERSITY

Team 56: OpenROAD VLSI Design 2 Status Update 3

Kevin, Talha, Githin, Giovanni
Sponsor: Kevin Nowka
TA: Michael Johnson-Moore

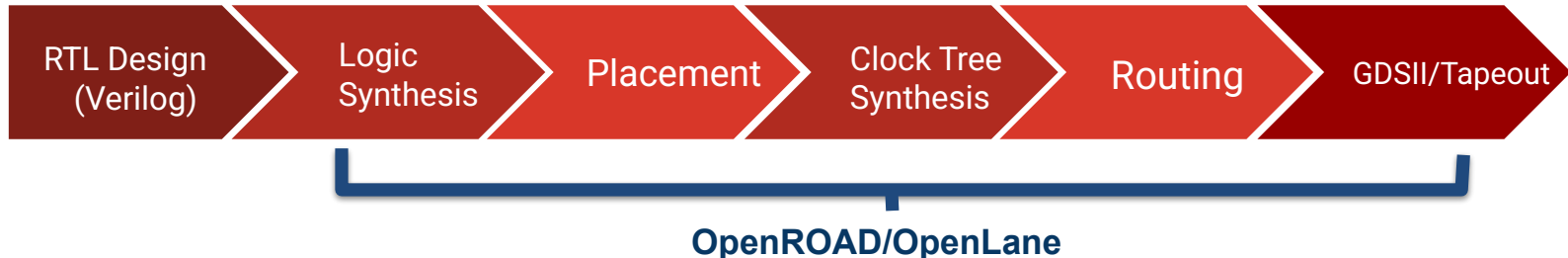
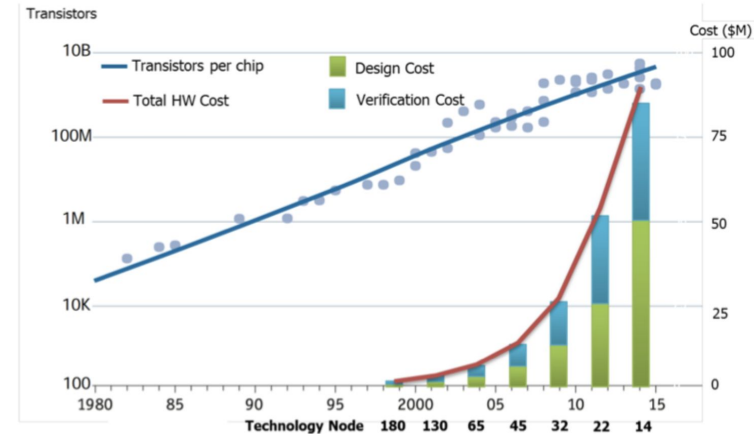
Project Summary

Problem statement:

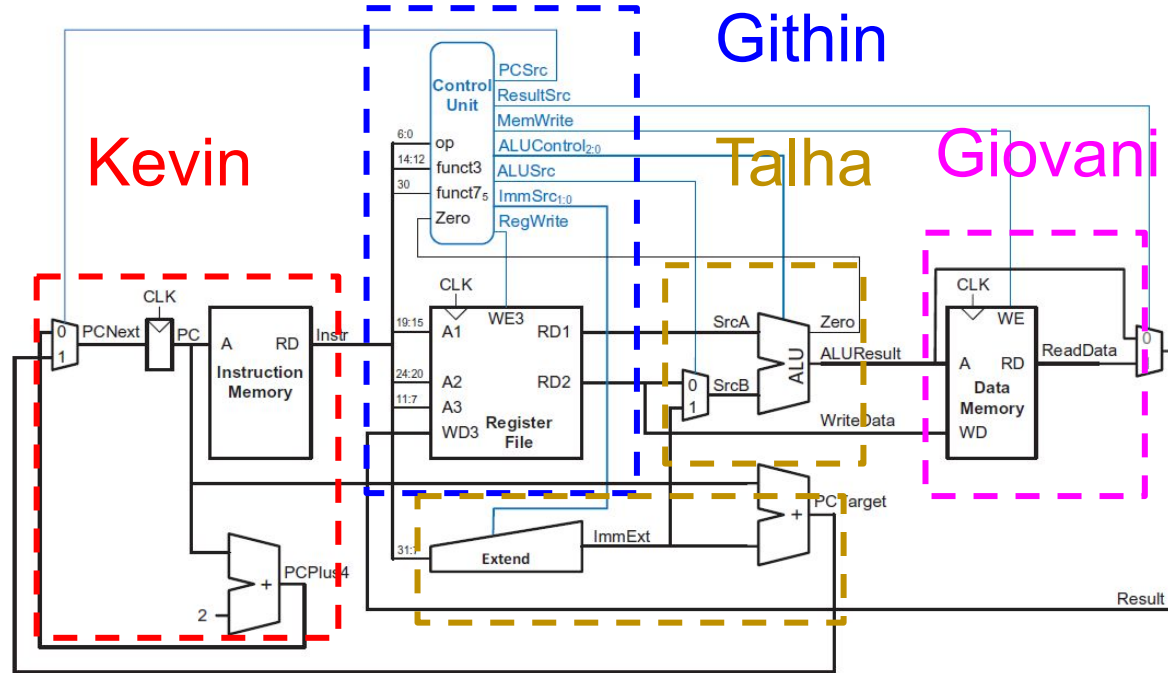
- Traditional RTL-to-GDSII chip design is expensive, creating a cost barrier to developing a System on a Chip (SoC).

Our Solution:

- Create a 16-bit RISC-V using OpenROAD, which automates chip design by only needing only an RTL description to generate GDSII, significantly reducing cost.



System Overview





Project Timeline

Finalize individual subsystem (completed 1/26)	Start integration/validation (Start by 2/2)	Implement pipeline stages (Start by 2/16)	Validate pipeline stages (Start by 3/2)	Start OpenLane/OpenR OAD process (Start by 3/23)	Final Validation (Start by 4/6)
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Fetch

Kevin Umanzor

Accomplishments since status update 2 15 hrs of effort	Ongoing progress/problems and plans until next presentation
- Finished implementing branch-type instructions by updating the ISA to ensure optimal bit usage. - Began researching I/O peripherals.	- Continuing research on pre-designed drivers and ESD protection - Determining the placement of I/O control logic and defining the I/O pad and pad ring architecture.

Fetch

Kevin Umanzor

```

=====
SECTION 7: BEQ - TAKEN / NOT TAKEN
=====
>>          Taken (5==5): x4 poison SKIPPED -> x4 stays intact -> S11=15
-----
[PASS] x4=15 (BEQ taken: poison skipped)                0x000f (15)

>>          Not-taken (5!=7): x5 sentinel RAN -> accumulates to S11=15
-----
[PASS] x5=15 (BEQ not-taken: sentinel executed)          0x000f (15)

>>          x0==x0 always taken: x6 poison SKIPPED -> S11=10
-----
[PASS] x6=10 (BEQ x0==x0: x6 poison skipped)             0x000a (10)

=====
SECTION 8: BNE - TAKEN / NOT TAKEN
=====
>>          Taken (3!=7): x3 poison SKIPPED -> x3 intact -> S11=30
-----
[PASS] x3=30 (BNE taken: x3 poison skipped)              0x001e (30)

>>          Not-taken (3==3): x4 sentinel RAN -> S11=15
-----
[PASS] x4=15 (BNE not-taken: sentinel executed)          0x000f (15)

>>          Taken (3!=0): x6 poison SKIPPED -> S11=10
-----
[PASS] x6=10 (BNE taken 3!=0: x6 poison skipped)        0x000a (10)

=====
SECTION 9: BLT - TAKEN / NOT TAKEN
=====
>>          Taken (2<5): x3 poison SKIPPED -> S11=30
-----
[PASS] x3=30 (BLT taken 2<5: x3 poison skipped)          0x001e (30)

>>          Not-taken (5 not<2): x4 sentinel RAN -> S11=15
-----
[PASS] x4=15 (BLT not-taken 5 not<2: sentinel executed)  0x000f (15)

>>          Taken (-1<2): x6 poison SKIPPED -> S11=10
-----
[PASS] x6=10 (BLT taken -1<2: x6 poison skipped)        0x000a (10)

=====
SECTION 10: BGE - TAKEN / NOT TAKEN
=====
>>          Taken (5>=2): x3 poison SKIPPED -> S11=30
-----
[PASS] x3=30 (BGE taken 5>=2: x3 poison skipped)        0x001e (30)

>>          Not-taken (2 not>=5): x4 sentinel RAN -> S11=15
-----
[PASS] x4=15 (BGE not-taken 2 not>=5: sentinel executed) 0x000f (15)

>>          Equal taken (5>=5): x5 poison SKIPPED -> S11=15
-----
[PASS] x5=15 (BGE equal-taken 5>=5: x5 poison skipped)  0x000f (15)

=====
RESULT: ALL 12 BRANCH TESTS PASSED
=====

```

Decode

Githin Johny

Accomplishments since status update 2 20 hrs of effort	Ongoing progress/problems and plans until next presentation
- Completed decode stage integration with the full processor - Successfully validated behavior of I-type instruction within the full processor	Continue to implement 3-stage pipeline into processor

I-type Instruction Validation

Githin Johny

0000
1481
0d01
0481
fd81
fa01
0d09
1d11
1519
7e81
8301

Sample Program

```
=====
RISCV-16 - I-Type Instruction Tests (10 cases)
=====
Time=16000 Reset released, PC=0x0000

--- Execution trace ---
Time | PC | Instr | ALU | x1 | x2 | x3 | x4 | x5 | x6
-----+-----+-----+-----+-----+-----+-----+-----+-----+-----
26000 | 0x0002 | 1481 | 0005 | 0000 | 0000 | 0000 | 0000 | 0000 | 0000
36000 | 0x0004 | 0d01 | 0003 | 0005 | 0000 | 0000 | 0000 | 0000 | 0000
46000 | 0x0006 | 0481 | 0006 | 0005 | 0003 | 0000 | 0000 | 0000 | 0000
56000 | 0x0008 | fd81 | ffff | 0006 | 0003 | 0000 | 0000 | 0000 | 0000
66000 | 0x000a | fa01 | fffe | 0006 | 0003 | ffff | 0000 | 0000 | 0000
76000 | 0x000c | 0d09 | 0000 | 0006 | 0003 | ffff | fffe | 0000 | 0000
86000 | 0x000e | 1d11 | 0007 | 0006 | 0000 | ffff | fffe | 0000 | 0000
96000 | 0x0010 | 1519 | 0005 | 0006 | 0007 | ffff | fffe | 0000 | 0000
106000 | 0x0012 | 7e81 | 001f | 0006 | 0005 | ffff | fffe | 0000 | 0000
116000 | 0x0014 | 8301 | ffe0 | 0006 | 0005 | ffff | fffe | 001f | 0000

--- Assertion checks ---
PASS      ADDI pos: x1 = 5+1 = 6 (got 0x0006)
PASS      ADDI/XORI/ORI/ANDI: x2 = 5 (got 0x0005)
PASS      Negative 1 - (-1) = 0xFFFF (got 0xffff)
PASS      Negative 2 - (-2) = 0xFFFE (got 0xfffe)
PASS      ADDI MAX value (31) (0x001F) (got 0x001f)
PASS      ADDI min value (-32) (0xFFE0) (got 0xffe0)
PASS      x7 untouched: x7 = 0x0000 (got 0x0000)

=====
All 7 / 7 tests PASSED.
=====
```




Execute

Talha Ibrahim

Accomplishments since status update 2 13 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Successfully integrated the Execute stage with the top level processor	• Further Validating the R-type instructions for the chip • Researching on I/O peripherals

Data Memory

Giovani Giagnacovo

Accomplishments since status update 2 9 hrs of effort	Ongoing progress/problems and plans until the next presentation
• Successful functionality of Data Memory subsystem in full CPU top level design for load and store instructions	• Expand upon functional top level design to ensure top level testbench is comprehensive for all instruction types. • Implementing pipelining and I/O implementation into our design.

Data Memory

Giovani Giagnacovo

--- Execution trace ---

Time	PC	Instr	ALU	x1	x2	x3	x4	x5	Note
26000	0x0002	1481	0005	0000	0000	0000	0000	0000	0000
36000	0x0004	0d01	0003	0005	0000	0000	0000	0000	0000
46000	0x0006	2980	0008	0005	0003	0000	0000	0000	0000
56000	0x0008	6a08	0005	0005	0003	0008	0000	0000	0000
66000	0x000a	2a98	0007	0005	0003	0008	0005	0000	0000
76000	0x000c	2b20	0001	0005	0003	0008	0005	0007	0000
86000	0x000e	2b90	0006	0005	0003	0008	0005	0007	0001
96000	0x0010	004e	0040	0005	0003	0008	0005	0007	0001
106000	0x0012	0d09	0000	0040	0003	0008	0005	0007	0001
116000	0x0014	1511	0005	0040	0000	0008	0005	0007	0001
126000	0x0016	1d19	0005	0040	0005	0008	0005	0007	0001
136000	0x0018	0413	0002	0040	0005	0008	0005	0007	0001
146000	0x001a	041a	0002	0040	0005	0008	0005	0007	0001
156000	0x001c	069c	0000	0040	0005	0005	0005	0007	0001
166000	0x0022	00a5	0002	0040	0005	0005	0005	0007	0001
176000	0x0026	8280	0002	0040	0005	0005	0002	0007	0001
186000	0x0028	0001	0000	0040	0005	0005	0002	0002	0001
196000	0x002a	0000	0000	0040	0005	0005	0002	0002	0001

PASS

L-type LW: x3 = mem[2] = 5 (written by SW x2,2(x0)) (got 0x0005)

PASS

S-type SW: mem[2] == 5 (byte addr 2 = word 1) (got 0x0005)



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Parts Ordering Status

Not Applicable

Execution Plan

[illegible]



Validation Plan

Paragraph #	Test Name	Success Criteria	Methodology	Status	Responsible Engineer(s)
3.1	Tile Size Constraint	The final GDSII layout must remain within the 2-tile maximum area allowed for TinyTapeout.	Run final Place and Route in OpenROAD, verify bounding box and padframe placement using the layout area report.	Untested	Full Team
4.1	DRC	Zero DRC violations (including antenna) in Magic and KLayout using the Sky130A/TinyTapeout rule decks.	Run Magic and KLayout DRC on the final GDS and iterate fixes until zero violations.	Untested	Full Team
4.2	Operating Temperature Range	Design must remain functionally valid across process and temperature corners (-55°C to +125°C).	Run timing and power analysis using PDK corner simulations (TT, SS, FF) to ensure performance remains stable within range.	Untested	Talha Ibrahim, Kevin Umanzor
5.1	LVS	Netgen reports "LVS clean" with 0 opens/shorts and all devices/nets/pins matched.	Extract the layout to SPICE and compare against the synthesized gate-level netlist in Netgen with the sky130A setup.	Untested	Full Team
5.2	Primary Input Power	Active power below 200 μ W and lower during idle conditions.	Estimate and verify power through post-layout power analysis and simulation reports in OpenROAD.	Untested	Talha Ibrahim, Kevin Umanzor
5.3	Signal Interfaces	All interfaces (I ² C, SPI, USB) respond correctly at specified logic levels and timings.	Run RTL and gate-level testbenches to verify correct data transfer, logic levels, and signal timing for all interfaces.	Untested	Githin Johny, Giovanni Giagnacovo
5.4	User Control Interface	The reset mechanism returns the CPU to a known initial state.	Run RTL simulations for both synchronous and asynchronous resets to confirm the PC and register file reset correctly.	Untested	Giovani Giagnacovo, Talha Ibrahim
5.5	Clock Interface	The design accepts a 10 MHz external clock and meets all setup and hold timing constraints.	Run Static Timing Analysis (STA) and post-layout timing simulations in OpenROAD to verify clock stability and timing closure.	Untested	Kevin Umanzor, Githin Johny
6.1	Fetch	The Fetch stage retrieves correct instructions and updates the program counter properly.	Create directed and random test programs in RTL simulation to verify instruction fetch and PC increment behavior.	Tested	Kevin Umanzor
6.2	Decode	The Decode stage correctly generates control signals from instruction fields.	Run RTL testbenches for all supported instructions to confirm control signal accuracy and complete coverage.	Tested	Githin Johny
6.3	Execute	The ALU produces correct results for all arithmetic and logic operations.	Run RTL and gate-level simulations for arithmetic, logical, and branch operations with directed and random test vectors.	Tested	Talha Ibrahim
6.4	Memory/Write Back	The memory and write-back stages perform correct data reads, writes, and register updates.	Run RTL and gate-level simulations with memory models to verify correct load/store behavior and data integrity.	Tested	Giovani Giagnacovo
	Full System Simulation	The full 16-bit RISC-V design executes test programs correctly through all pipeline stages.	Integrate all modules into a top-level RTL and post-layout simulation, verifying instruction execution, output correctness, and timing.	Untested	Full Team
	Full System Demo	Execute a test program on each corresponding part on the chip measuring correct output for all cases; along with measured power and basic I/O pass	Integrate tests for top level regression and postlayout simulation capturing outputs.	Untested	Full Team



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Thank you!!